THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student argues that CO2 is bad and should be removed entirely. A good reply is: 1

(a) CO2 has no role
(b) CO2 is not a greenhouse gas
(c) CO2 is made by plants only
(d) without CO2 the Earth would be too cold and plants could not make food

2. Assertion (A): Weather occurs mostly in the troposphere. Reason (R): The troposphere contains most of the water vapour of the atmosphere. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Which of the following correctly orders the layers by temperature trend (down to up) in the lower atmosphere? 1

(a) decreasing, increasing, decreasing
(b) increasing, decreasing, increasing
(c) all increasing
(d) all decreasing

4. Fresh water from desalination is expensive mainly because: 1

(a) water is heavy
(b) large energy is needed to remove salt
(c) salt is rare
(d) sea is far

5. Assertion (A): Seismic waves help in studying the interior of the Earth. Reason (R): Waves change speed on passing through materials of different density. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Mars has seasons similar to Earth because: 1

(a) it has a Moon
(b) its axis is tilted similarly
(c) it has a thick atmosphere
(d) it has oceans

7. Migratory birds find direction using the: 1

(a) ozone layer
(b) Earth's magnetic field
(c) wind only
(d) Moon only

8. Assertion (A): The sky appears black when seen from the Moon. Reason (R): The Moon has no atmosphere to scatter sunlight. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. If the Earth were twice as far from the Sun, the likely effect would be: 1

(a) freezing of most water **(b)** no change
(c) more rain **(d)** boiling of oceans

10. The Earth's biosphere extends: 1

(a) to the Moon **(b)** to the core
(c) only to the crust **(d)** from deep oceans to the lower atmosphere

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is exploring Mars and other planets helpful in understanding Earth? 2

12. Why are the polar regions more affected by global warming? 2

13. Why does the sea level rise when land ice melts but not when floating ice melts? 2

14. What is the 'carbon footprint' and why should we reduce it? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Compare the stratospheric ozone and ground-level ozone in terms of origin and effect. 3

16. A classroom debates 'Should we explore other planets or fix the Earth first?' Present two points on each side. 3

17. Explain how the Earth's tilt would affect the climate if the tilt were zero. 3

OR

Explain the changes in air pressure and temperature with height in the troposphere. 3

18. Explain why some countries are more affected by sea-level rise and what adaptation measures can be taken. 3

OR

Explain the need to protect the Earth as 'our only home' with three arguments. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe how human activities have changed the composition of the atmosphere in the last 200 years and the consequences for ecosystems and societies. 5

OR

A student claims that 'the Earth is so large that human activities cannot change it'. Evaluate the claim with three pieces of evidence from the atmosphere, hydrosphere and biosphere. 5

20. Explain, with reasoning, how liquid water became stable on Earth and what would happen if the Earth were 10 per cent closer or 10 per cent farther from the Sun. 5

OR

Explain how the Earth's systems (atmosphere, hydrosphere, lithosphere, biosphere) keep a balance and what happens if one is disturbed, using the carbon cycle as an example. 5

END OF QUESTION PAPER — Total: 40 Marks